2025/09 PHY301 Thermodynamic and Statistical Physics

Problem set 3

Quantum statistics, Fermi and Bose gas

Problem 1: Density matrix for two-particle system[25 marks]

Consider a single harmonic oscillator with the Hamiltonian

$$\mathcal{H} = -\frac{\hbar^2}{2m}\frac{\partial}{\partial^2 x^2} + \frac{m\omega^2}{2}x^2$$

with eigenvalues $\varepsilon_n = (n + \frac{1}{2})\hbar\omega$ and eigenfunction

$$\phi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{H_n(\xi)}{(2^n n!)^{1/2}} \mathrm{e}^{-\xi^2/2}$$

where $\xi = (m\omega/\hbar)^{1/2}x$. The Hermite polynomial can be obtained by

$$H_n(\xi) = (-1)^n \mathrm{e}^{\xi^2} \frac{\partial^n}{\partial \xi^n} \mathrm{e}^{-\xi^2}$$

In canonical ensemble, the density matrix is $\rho = \mathrm{e}^{-\beta\mathcal{H}}/\mathcal{Z}$ where the partition function $\mathcal{Z} = \mathrm{tr}\mathrm{e}^{-\beta\mathcal{H}}$.

(a) Show that the density matrix in real space x-representation is

$$\langle x|\mathrm{e}^{-\beta\mathcal{H}}|x'\rangle = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \mathrm{e}^{-\frac{1}{2}(\xi^2+\xi'^2)} \sum_{n=0}^{\infty} \frac{\mathrm{e}^{-\beta(n+1/2)\hbar\omega}}{2^n n!} H_n(\xi) H_n(\xi')$$

Solution

Using the completeness, $1 = \sum_n = 0^{\infty} |n\rangle\langle n|$ where $\mathcal{H}|n\rangle = \varepsilon_n|n\rangle$ is the energy eigenstate. Therefore, we have

$$\begin{aligned}\langle x|\mathrm{e}^{-\beta\mathcal{H}}|x'\rangle &= \sum_{n=0}^{\infty}\langle x\mathrm{e}^{-\beta\mathcal{H}}|n\rangle\langle n|x'\rangle \\ &= \sum_{n=0}^{\infty}\mathrm{e}^{-\beta\varepsilon_n}\langle x|n\rangle\langle n|x'\rangle\end{aligned}$$

Using $\langle x|n\rangle = \phi_n(x)$, we have

$$\langle x|\mathrm{e}^{-\beta\mathcal{H}}|x'\rangle = \sum_{n=0}^{\infty}\mathrm{e}^{-\beta\varepsilon_n}\phi_n(x)\phi_n(x')$$

Using $\varepsilon_n = (n+\frac{1}{2})\hbar\omega$ and the eigenfunction in the question, we, therefore, obtain the desirable result. **[5 marks]**

(b) To evaluate the about n-sum, we can use a very useful and common trick to turn the derivative into a product of polynomials by using Fourier transformation. Namely,

$$\frac{d^n}{dx^n}f(x) = \int_{-\infty}^{\infty}(ik)^n\mathrm{e}^{ikx}\tilde{f}(k)\frac{dk}{2\pi}$$

where $\tilde{f}(k) = \int_{-\infty}^{\infty} f(x)\mathrm{e}^{-ikx}$ is the fourier transformation of $f(x)$. With this, prove the following

$$\begin{aligned}&\sum_{n=0}^{\infty}\frac{\mathrm{e}^{-\beta(n+1/2)\hbar\omega}}{2^n n!}H_n(\xi)H_n(\xi') \\ &\quad= \frac{\mathrm{e}^{-\frac{1}{2}\beta\hbar\omega}\mathrm{e}^{\xi^2+\xi'^2}}{4\pi}\int dk\int dk'\exp[-\frac{1}{4}(k^2+2kk'\mathrm{e}^{-\beta\hbar\omega}+k'^2)+i(k\xi+k'\xi')]\end{aligned} \tag{1}$$

Solution

Using Fourier transformation, we write

$$H_n(\xi) = (-1)^n \mathrm{e}^{\xi^2} \int_{-\infty}^{\infty} (ik)^n \mathrm{e}^{ik\xi} \mathcal{F}[\mathrm{e}^{-\xi^2}]$$

where the Fourier transformation of the Gaussian function is

$$\mathcal{F}[\mathrm{e}^{-\xi^2}] = \int_{-\infty}^{\infty} d\xi \mathrm{e}^{-ik\xi} \mathrm{e}^{-\xi^2} = \sqrt{\pi} \mathrm{e}^{-k^2/4}$$

Therefore, we can write

$$\begin{aligned}
\sum_{n=0}^{\infty} \frac{\mathrm{e}^{-\beta(n+1/2)\hbar\omega}}{2^n n!} H_n(\xi) H_n(\xi') = & \sum_{n=0}^{\infty} \frac{\mathrm{e}^{-\beta(n+1/2)\hbar\omega}}{2^n n!} \int \frac{dk}{2\pi} \int \frac{dk'}{2\pi} \\
& \mathrm{e}^{\xi^2+\xi'^2} (-kk')^n \pi \mathrm{e}^{i(k\xi+k'\xi')} \mathrm{e}^{-(k^2+k'^2)/4} \\
= & \mathrm{e}^{-\beta\hbar\omega/2} \iint \frac{dkdk'}{4\pi} \sum_{n=0}^{\infty} \frac{\mathrm{e}^{\xi^2+\xi'^2}}{n!} (-\tfrac{1}{2} kk' \mathrm{e}^{-\beta\hbar\omega})^n \\
& \mathrm{e}^{i(k\xi+k'\xi')} \mathrm{e}^{-(k^2+k'^2)/4} \\
= & \mathrm{e}^{-\beta\hbar\omega/2} \iint \frac{dkdk'}{4\pi} \mathrm{e}^{\xi^2+\xi'^2} \exp\Big[-\tfrac{1}{2} kk' \mathrm{e}^{-\beta\hbar\omega} \\
& + i(k\xi + k'\xi') - \tfrac{1}{4}(k^2 + k'^2) \Big]
\end{aligned}$$

Further simplifying the above, we can therefore obtain the result in (b). **[5 marks]**

(c) The above integration is the *generalized* version of Gaussian integral in higher dimensional vector space. For instance, in a n-dimensional vector space,

$$\int dx^n \mathrm{e}^{-\frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x}+\mathbf{x}^T\mathbf{J}} = \sqrt{\frac{(2\pi)^n}{|\mathbf{A}|}} \mathrm{e}^{-\frac{1}{2}\mathbf{J}^T\mathbf{A}^{-1}\mathbf{J}} \tag{2}$$

where the $n \times n$ matrix $[\mathbf{A}]_{ij} = A_{ij}$, the vectors $\mathbf{x} = [x_1, \ldots x_n]$ and $\mathbf{J} = [J_1, \ldots, J_n]$. [*Note.* Gaussian integration in n-dimensional vector space is an important integration that we need to know in physics. You can derive or find the proof by yourself.]. Using Eq.(2), we therefore integrat out k and k' in Eq.(1), and show that the density matrix

is

$$\langle x|\mathrm{e}^{-\beta\mathcal{H}}|x'\rangle = \sqrt{\frac{m\omega}{2\pi\hbar\sinh(\beta\hbar\omega)}}\exp\Big[-\frac{m\omega}{4\hbar}\Big((x+x')^2\tanh\frac{\beta\hbar\omega}{2}+(x-x')^2\coth\frac{\beta\hbar\omega}{2}\Big)\Big]$$

Solution

To integrate out k and k' in (b), we have first identify

$$\mathbf{A} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2}\mathrm{e}^{-\beta\hbar\omega} \\ \frac{1}{2}\mathrm{e}^{-\beta\hbar\omega} & \frac{1}{2} \end{bmatrix}, \quad \mathbf{J}^T = [i\xi, i\xi'], \quad \text{, and} \quad \mathbf{x} = [k, k']$$

We then calculate

$$|\mathbf{A}| = \frac{1}{4}(1-\mathrm{e}^{-2\beta\hbar\omega}), \quad \mathbf{A}^{-1} = \frac{4}{1-\mathrm{e}^{-2\beta\hbar\omega}}\begin{bmatrix} \frac{1}{2} & -\frac{1}{2}\mathrm{e}^{-\beta\hbar\omega} \\ -\frac{1}{2}\mathrm{e}^{-\beta\hbar\omega} & \frac{1}{2} \end{bmatrix}$$

and

$$\frac{1}{2}\mathbf{J}^T\mathbf{A}^{-1}\mathbf{J} = -\frac{2}{1-\mathrm{e}^{-2\beta\hbar\omega}}[\xi,\xi']\begin{bmatrix} \frac{1}{2} & -\frac{1}{2}\mathrm{e}^{-\beta\hbar\omega} \\ -\frac{1}{2}\mathrm{e}^{-\beta\hbar\omega} & \frac{1}{2} \end{bmatrix}\begin{bmatrix} \xi \\ \xi' \end{bmatrix} = -\frac{\xi^2-2\xi\xi'\mathrm{e}^{-\beta\hbar\omega}+\xi'^2}{1-\mathrm{e}^{-2\beta\hbar\omega}}$$

Therefore, we find

$$\sum_{n=0}^{\infty}\frac{\mathrm{e}^{-\beta(n+1/2)\hbar\omega}}{2^n n!}H_n(\xi)H_n(\xi') = \frac{\mathrm{e}^{-\frac{1}{2}\beta\hbar\omega}\mathrm{e}^{\xi^2+\xi'^2}}{4\pi}\sqrt{\frac{4\times 2^2\pi^2}{1-\mathrm{e}^{-2\beta\hbar\omega}}}\exp\Big[-\frac{\xi^2-2\xi\xi'\mathrm{e}^{-\beta\hbar\omega}+\xi'^2}{1-\mathrm{e}^{-2\beta\hbar\omega}}\Big]$$

This leads to

$$\langle x|\mathrm{e}^{-\beta\mathcal{H}}|x'\rangle = \Big(\frac{m\omega}{\pi\hbar}\Big)^{1/2}\frac{\exp\Big[\frac{1}{2}(\xi^2+\xi'^2)\Big]}{\sqrt{2\sinh(\beta\hbar\omega)}}\exp\Big[-\frac{\xi^2\mathrm{e}^{\beta\hbar\omega}-2\xi\xi'+\xi'^2\mathrm{e}^{\beta\hbar\omega}}{\mathrm{e}^{\beta\hbar\omega}-\mathrm{e}^{-\beta\hbar\omega}}\Big]$$

We further simplify the above equation

$$\begin{aligned}\langle x|\mathrm{e}^{-\beta\mathcal{H}}|x'\rangle &= \Big(\frac{m\omega}{2\pi\hbar\sinh(\beta\hbar\omega)}\Big)^{1/2}\exp\Big[\frac{-\xi^2(\mathrm{e}^{\beta\hbar\omega}+\mathrm{e}^{-\beta\hbar\omega})+4\xi\xi'-\xi'^2(\mathrm{e}^{\beta\hbar\omega}+\mathrm{e}^{-\beta\hbar\omega})}{2(\mathrm{e}^{\beta\hbar\omega}-\mathrm{e}^{-\beta\hbar\omega})}\Big] \\ &= \Big(\frac{m\omega}{2\pi\hbar\sinh(\beta\hbar\omega)}\Big)^{1/2}\exp\Big[\frac{-2\xi^2\cosh(\beta\hbar\omega)+4\xi\xi'-2\xi'^2\cosh(\beta\hbar\omega)}{4\sinh(\beta\hbar\omega)}\Big]\end{aligned}$$

Solution

Using the identities

$$\sinh 2x = 2\sinh x \cosh x$$

$$\cosh 2x = \sinh^2 x + \cosh^2 x$$

$$\cosh^2 x - \sinh^2 x = 1$$

We, therefore, obtain

$$\langle x|\mathrm{e}^{-\beta\mathcal{H}}|x'\rangle = \Big(\frac{m\omega}{2\pi\hbar\sinh(\beta\hbar\omega)}\Big)^{1/2} \exp\Big[\frac{-2(\xi-\xi')^2\cosh^2(\beta\hbar\omega/2) - 2(\xi+\xi')\sinh^2(\beta\hbar\omega/2)}{8\sinh(\beta\hbar\omega/2)\cosh(\beta\hbar\omega/2)}\Big]$$

Further simplify the above, it reduces to the result in (c). **[5 marks]**

(d) Calculate the partition function and density matrix

Solution

The partition function is

$$\mathcal{Z} = \mathrm{tr}\mathrm{e}^{-\beta\mathcal{H}} = \int dx\langle x|\mathrm{e}^{-\beta\mathcal{H}}|x\rangle = \int dx\sqrt{\frac{m\omega}{2\pi\hbar\sinh(\beta\hbar\omega)}}\exp\Big[-\frac{m\omega}{\hbar}\tanh\frac{\beta\hbar\omega}{2}x^2\Big]$$

Therefore, we obtain

$$\mathcal{Z} = \sqrt{\frac{m\omega}{2\pi\hbar\sinh(\beta\hbar\omega)}}\sqrt{\frac{\pi\hbar}{m\omega\tanh\frac{\beta\hbar\omega}{2}}} = \frac{1}{2\sinh\frac{\beta\hbar\omega}{2}}$$

This gives the density matrix

$$\rho = \frac{\langle x|\mathrm{e}^{-\beta\mathcal{H}}|x'\rangle}{\mathcal{Z}} = \sqrt{\frac{m\omega\tanh(\beta\hbar\omega/2)}{\pi\hbar}}\exp\Big[-\frac{m\omega}{4\hbar}\Big((x+x')^2\tanh\frac{\beta\hbar\omega}{2} + (x-x')^2\coth\frac{\beta\hbar\omega}{2}\Big)$$

[5 marks]

(e) Using the density matrix, calculate $\langle x^2\rangle$

Solution

$$\langle x^2\rangle = \mathrm{tr}x^2\rho = \int x^2 \frac{\langle x|\mathrm{e}^{-\beta\mathcal{H}}|x\rangle}{\mathcal{Z}}dx = \int x^2 \sqrt{\frac{m\omega\tanh(\beta\hbar\omega/2)}{\pi\hbar}}\exp\left[-\frac{m\omega}{\hbar}x^2\tanh\frac{\beta\hbar\omega}{2}\right]dx$$

Using

$$\int x^2\mathrm{e}^{-ax^2} = \sqrt{\frac{\pi}{a}}\frac{1}{2a}$$

this leads to

$$\langle x^2\rangle = \sqrt{\frac{m\omega\tanh(\beta\hbar\omega/2)}{\pi\hbar}}\sqrt{\frac{\pi}{\frac{m\omega}{\hbar}\tanh\frac{\beta\hbar\omega}{2}}}\frac{1}{2\frac{m\omega}{\hbar}\tanh\frac{\beta\hbar\omega}{2}} = \frac{\hbar}{2m\omega\tanh\frac{\beta\hbar\omega}{2}}$$

[3 marks]

(f) Using the density matrix, calculate $E = \langle H\rangle$.

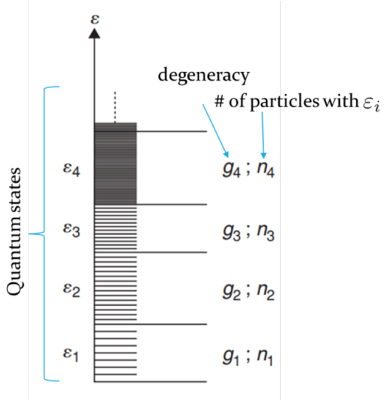
Solution

$$E = \mathrm{tr}\mathcal{H}\rho = -\frac{\partial\ln\mathcal{Z}}{\partial\beta} = \frac{\partial\ln(2\sinh\frac{\beta\hbar\omega}{2})}{\partial\beta} = \frac{\hbar\omega}{2}\coth\frac{\beta\hbar\omega}{2}$$

[2 marks]

Problem 2: Quantum statistics [25 marks]

The one-particle's energy spectrum is demonstrated in the figure below.



The N-particle Hamiltonian is

$$\mathcal{H}_N(\{n_{i\alpha}\}) = \sum_{i=1} \sum_{\alpha=1}^{g_i} \varepsilon_i n_{i\alpha} \tag{3}$$

Here, $n_{i\alpha}$ is the occupation number of particles in state α with energy ε_i and $n_i = \sum_{\alpha=1}^{g_i} n_{i\alpha}$. In canonical ensemble with N identical particles, the partition function is $\mathcal{Z}_N = \sum_{\{n_{i\alpha}\}} \mathrm{e}^{-\beta \mathcal{H}_N(\{n_{i\alpha}\})}$ The sum is subject to the constraint of total number of particles $\sum_i \sum_{\alpha=1}^{g_i} n_{i\alpha} = N$. To relax this constraint, we use grand canonical ensemble to model our quantum system and the grand partition

$$\mathcal{Q} = \sum_{N=0}^{\infty} \mathrm{e}^{\beta \mu N} \mathcal{Z}_N$$

(a) Calculate the Grand partition function for *Boson* and *Fermion*.

Solution

The Grand partition function

$$\mathcal{Q} = \sum_{N=0}^{\infty} \mathrm{e}^{\beta\mu N} \sum_{\{n_{i\alpha}\}} \mathrm{e}^{-\beta \sum_{i=1} \sum_{\alpha=1}^{g_i} \varepsilon_i n_{i\alpha}} = \sum_{N=0}^{\infty} \sum_{\{n_{i\alpha}\}} \mathrm{e}^{-\beta \sum_{i=1} \sum_{\alpha=1}^{g_i} (\varepsilon_i - \mu) n_{i\alpha}}$$

For Boson, we have

$$\begin{aligned}
\mathcal{Q} &= \Big(\sum_{n_{1,1}=0}^{\infty} \cdots \sum_{n_{1,g_1}=0}^{\infty} \Big) \Big(\sum_{n_{2,1}=0}^{\infty} \cdots \sum_{n_{2,g_2}=0}^{\infty} \Big) \cdots \prod_{i=1} \prod_{\alpha=1}^{g_i} \mathrm{e}^{-\beta(\varepsilon_i - \mu) n_{i\alpha}} \\
&= \prod_{i=1} \prod_{\alpha=1}^{g_i} \sum_{n_{i\alpha}=0}^{\infty} \mathrm{e}^{-\beta(\varepsilon_i - \mu) n_{i\alpha}} \\
&= \prod_{i=1} \Big(\frac{1}{1 - \mathrm{e}^{-\beta(\varepsilon_i - \mu)}} \Big)^{g_i}
\end{aligned}$$

For Fermion, we have

$$\begin{aligned}
\mathcal{Q} &= \Big(\sum_{n_{1,1}=0}^{1} \cdots \sum_{n_{1,g_1}=0}^{1} \Big) \Big(\sum_{n_{2,1}=0}^{1} \cdots \sum_{n_{2,g_2}=0}^{1} \Big) \cdots \prod_{i=1} \prod_{\alpha=1}^{g_i} \mathrm{e}^{-\beta(\varepsilon_i - \mu) n_{i\alpha}} \\
&= \prod_{i=1} \prod_{\alpha=1}^{g_i} \sum_{n_{i\alpha}=0}^{1} \mathrm{e}^{-\beta(\varepsilon_i - \mu) n_{i\alpha}} \\
&= \prod_{i=1} \Big(1 + \mathrm{e}^{-\beta(\varepsilon_i - \mu)} \Big)^{g_i}
\end{aligned}$$

[9 marks]

(b) Using the result in (a), calculate the mean occupation number $\langle n_i \rangle = \langle \sum_{\alpha_i=1}^{g_i} n_{i\alpha} \rangle$ for both bosonic and fermionic system

Solution

The mean occupation number can be calculated by using

$$\langle n_{i\alpha} \rangle = -\frac{1}{\beta} \frac{\partial \ln \mathcal{Q}}{\partial \varepsilon_i}$$

For Boson, we have

$$\langle n_{i\alpha} \rangle = -\frac{1}{\beta} \frac{\partial}{\partial \varepsilon_i} \sum_{i'=1} g_{i'} \ln \frac{1}{1 - \mathrm{e}^{-\beta(\varepsilon_{i'} - \mu)}} = \frac{g_i}{\mathrm{e}^{\beta(\varepsilon_i - \mu)} - 1}$$

For Fermion,

$$\langle n_{i\alpha} \rangle = -\frac{1}{\beta} \frac{\partial}{\partial \varepsilon_i} \sum_{i'=1} g_{i'} \ln(1 + \mathrm{e}^{-\beta(\varepsilon_{i'} - \mu)}) = \frac{g_i}{\mathrm{e}^{\beta(\varepsilon_i - \mu)} + 1}$$

[9 marks]

(c) What is the probability of finding $n_{i\alpha}$ particles in state α with energy ε_i.

Solution

The probability of find the system at microstates $\{n_{i\alpha}\}$ is

$$p(\{n_{i\alpha}\}) = \mathrm{e}^{-\beta\sum_i\sum_\alpha^{g_i}\mu n_{i\alpha}}\mathrm{e}^{-\beta\mathcal{H}(\{n_{i\alpha}\})}/\mathcal{Q}$$

To find the probability of state α with energy ε_i occupy by $n_{i\alpha}$ particle is

$$p(n_{i\alpha}) = \underbrace{\Big(\sum_{n_{1,1}=0}\cdots\sum_{n_{1,g_1}=0}\Big)\Big(\sum_{n_{2,1}=0}\cdots\sum_{n_{2,g_2}=0}\Big)\ldots}_{\text{exclude the sum of } n_{i\alpha}} p(\{n_{i\alpha}\})$$

Therefore, for boson

$$p(n_{i\alpha}) = \mathrm{e}^{-\beta(\varepsilon_i-\mu)n_{i\alpha}}(1-\mathrm{e}^{-\beta(\varepsilon_i-\mu)})$$

For Fermion,

$$p(n_{i\alpha}) = \frac{\mathrm{e}^{-\beta(\varepsilon_i-\mu)n_{i\alpha}}}{1+\mathrm{e}^{-\beta(\varepsilon_i-\mu)}}$$

[7 marks]

Problem 3: Bose Gas with arbitrary dispersion in d-dimension [25 marks]

consider a gas of non-interacting (spinless) bosons with energy $\varepsilon = cp$ ($p = |\mathbf{p}|$), contained in a box of volume $V = L^3$.

(a) Calculate the grand potential $\mathcal{G} = -k_BT\ln\mathcal{Q}$, and the density $n = N/V$, at a chemical potential μ. Express your answers in terms of Bose-Einstein function $g_\nu(z)$, where $z = \mathrm{e}^{\beta\mu}$, and

$$g_\nu(z) = \frac{1}{(\nu-1)!}\int_0^\infty \frac{x^{\nu-1}dx}{z^{-1}\mathrm{e}^x - 1}$$

Solution

The Grand partition function is

$$\mathcal{Q} = \prod_i \left(\frac{1}{1 - z\mathrm{e}^{-\beta\varepsilon_i}} \right)^{g_i} \tag{4}$$

Then, we let $g_i = 1$. The grand partition is

$$\mathcal{G} = k_B T \sum_i \ln(1 - z\mathrm{e}^{-\beta\varepsilon_i})$$

Using $\sum_i = \sum_k = V \int \frac{d^3k}{(2\pi)^3}$ This gives

$$\mathcal{G} = k_B T V \int \frac{d^3k}{(2\pi)^3} \ln(1 - z\mathrm{e}^{-\beta\varepsilon_i})$$

We change of variable $\varepsilon = \hbar c k$. We then have $\hbar c dk = d\varepsilon$.

$$\mathcal{G} = k_B T V \int \frac{\varepsilon^2}{2(\hbar c \pi)^2} \frac{d\varepsilon}{\hbar c} \ln(1 - z\mathrm{e}^{-\beta\varepsilon_i})$$

We further let $x = \beta\varepsilon$. This gives

$$\mathcal{G} = k_B T V \int \frac{x^2}{2\pi^2(\beta\hbar c)^3} \ln(1 - z\mathrm{e}^{-\beta\varepsilon_i}) dx$$

Integrating by part, we then obtain

$$\begin{aligned}
\mathcal{G} &= -\frac{k_B T V}{2\pi^2(\beta\hbar c)^3} \int \frac{x^3}{3} \frac{z\mathrm{e}^{-x}}{1 - z\mathrm{e}^{-\beta\varepsilon_i}} dx \\
&= -\frac{k_B T V}{\pi^2(\beta\hbar c)^3} \frac{1}{(4-1)!} \int \frac{x^{4-1}}{z^{-1}\mathrm{e}^{x} - 1} dx \\
&= -\frac{k_B T V}{\pi^2(\beta\hbar c)^3} g_4(z)
\end{aligned}$$

[6 marks]

(b) Calculate the gas pressure P, its energy U, and compare the ratio U/PV to the classical value.

Solution

Using $\mathcal{G} = -PV$. We find

$$P = \frac{k_B T}{\pi^2 (\beta \hbar c)^3} g_4(z)$$

The energy is

$$U = -\left(\frac{\partial}{\partial \beta} \ln \mathcal{Q}\right)_z = \left(\frac{\partial}{\partial \beta} \frac{\mathcal{G}}{k_B T}\right)_z$$

This gives

$$U = -\left(\frac{\partial}{\partial \beta} \frac{V}{\pi^2 (\beta \hbar c)^3} g_4(z)\right)_z = 3 \frac{V k_B T}{\pi^2 (\beta \hbar c)^3} g_4(z)$$

Therefore, we obtain

$$P/U = \frac{1}{3V} \Rightarrow PV/U = \frac{1}{3}$$

[6 marks]

(c) Find the critical temperature, $T_c(n)$, for Bose–Einstein condensation.

Solution

To calculate the transition temperature, we first evaluate the total number of particle

$$N = \sum_i \langle n_i \rangle = \sum_i \frac{1}{z^{-1}\mathrm{e}^{\beta\varepsilon_i} - 1} + \frac{1}{z^{-1} - 1}$$

Let $\sum_i \to \sum_k = V \int \frac{d^3k}{(2\pi)^3}$ and $N_0 = \frac{1}{z^{-1}-1}$,

$$\begin{aligned} N =& V \int \frac{d^3k}{(2\pi)^3} \frac{1}{z^{-1}\mathrm{e}^{\beta\varepsilon_k} - 1} + N_0 \\ =& V \int \frac{dk}{2\pi^2} \frac{k^2}{z^{-1}\mathrm{e}^{\beta\varepsilon_k} - 1} + N_0 \\ =& \frac{V}{(\hbar c)^3} \int \frac{d\varepsilon}{2\pi^2} \frac{\varepsilon^2}{z^{-1}\mathrm{e}^{\beta\varepsilon_k} - 1} + N_0 \end{aligned}$$

we let $x = \beta\varepsilon$,

$$\begin{aligned} N =& \frac{V}{\pi^2(\beta\hbar c)^3} \frac{1}{2} \int \frac{x^2 dx}{z^{-1}\mathrm{e}^{x} - 1} + N_0 \\ =& \frac{V}{\pi^2(\beta\hbar c)^3} g_3(z) + N_0 \end{aligned}$$

At the transition point $z = 1$ and $N_0 = 0$. This gives

$$N = \frac{V}{\pi^2(\beta\hbar c)^3} g_3(1) \Rightarrow N = \frac{V k_B^3 T^3}{\pi^2(\hbar c)^3} \zeta(3)$$

Therefore, the critical temperature

$$T_C = \left[\frac{V/N k_B^3}{\pi^2(\hbar c)^3} \zeta(3) \right]^{-1/3}$$

[6 marks]

(d) What is the temperature dependence of the heat capacity C_V for $T < T_c(n)$?

Solution

$$C_V = \frac{\partial U}{\partial T} = \frac{\partial}{\partial T}\frac{3Vk_B^4T^4}{\pi^2(\hbar c)^3}g_4(1) = \frac{12Vk_B^4T^3}{\pi^2(\hbar c)^3}\zeta(4) \propto T^3$$

[3 marks]

(e) Evaluate the dimensionless heat capacity C_V/Nk_B at the critical temperature $T = T_c$, and compare its value to the classical (high-temperature) limit.

Solution

$$\frac{C_V}{Nk_B} = \frac{12V/Nk_B^3}{\pi^2(\hbar c)^3}\zeta(4)T_C^3 = \frac{12V/Nk_B^3}{\pi^2(\hbar c)^3}\zeta(4)\left[\frac{V/Nk_B^3}{\pi^2(\hbar c)^3}\zeta(3)\right]^{-1} = 12\frac{\zeta(4)}{\zeta(3)}$$

$$\frac{C_V}{Nk_B} = 3$$

You can compare this with the classical result. **[4 marks]**

Problem 4: Non-interacting fermions [25 marks]

Consider a grand canonical ensemble of non-interacting fermions in the two-dimension crystal with chemical potential μ. The one-particle states are labeled by a wave vector $\vec{k}$, and have energies $\varepsilon_{\vec{k}} = \hbar^2k^2/(2m)$.

(a) What is the Fermi energy ε_F compare to 3D case?

Solution

$$\sum_k g = N$$
$$A \int \frac{d^2k}{(2\pi)^2} g = N$$
$$\frac{A}{2\pi} \int k dk g = N$$

where A is the area of the system. We change variable $k = \sqrt{2m\varepsilon}/\hbar$. This gives $dk = \frac{1}{2}\sqrt{2m}/\hbar\varepsilon^{-1/2}d\varepsilon$

$$A/(2\pi) \int_0^{\varepsilon_F} \frac{\sqrt{2m}}{\hbar} g\varepsilon^{1/2} (\tfrac{1}{2}\sqrt{2m}/\hbar\varepsilon^{-1/2}) d\varepsilon = N$$

Therefore, we have

$$gm/\hbar^2\varepsilon_F = 2\pi N/A \Rightarrow \varepsilon_F = \frac{2\pi\hbar^2}{gm}\frac{N}{A}$$

[5 marks]

(b) What is the chemical potential $\mu(T)$? [in this case you can find the exact expression for $\mu(T)$.]

Solution

We calculate the average total number

$$N = \sum_i \langle n_i \rangle = \sum_i \frac{g_i}{z^{-1}\beta\varepsilon_{\mathrm{i}} + 1}$$

we let $g_i = g$ and $\sum_i = \sum_k = A \int d^2k/(2\pi)^2$. This leads to

$$\begin{aligned} N =& A \int \frac{d^2k}{(2\pi)^2} \frac{g}{z^{-1}\mathrm{e}^{\beta\varepsilon_k j} + 1} \\ =& \frac{A}{2\pi} \int_0^\infty d\varepsilon \frac{gm/\hbar^2}{z^{-1}\mathrm{e}^{\beta\varepsilon_k} + 1} \end{aligned}$$

We can do integration exactly by letting $x = z^{-1}\mathrm{e}^{\beta\varepsilon}$. This gives $dx = \beta z^{-1}\mathrm{e}^{\beta\varepsilon} d\varepsilon$. Therefore,

$$\begin{aligned} N =& \frac{A}{2\pi\beta} \int_{z^{-1}}^\infty dx \frac{gm/\hbar^2}{x(x+1)} \\ \beta\varepsilon_F =& \int_{z^{-1}}^\infty dx \left(\frac{1}{x} - \frac{1}{x+1} \right) \\ \beta\varepsilon_F =& \left[\ln \frac{x}{x+1} \right]_{z^{-1}}^\infty \\ \beta\varepsilon_F =& -\ln \frac{z^{-1}}{z^{-1}+1} \\ \mathrm{e}^{-\beta\varepsilon_F} =& \frac{1}{z+1} \\ z =& \mathrm{e}^{-\beta\varepsilon_F} - 1 \end{aligned}$$

We then arrive at

$$\mu = \frac{1}{\beta} \ln(\mathrm{e}^{\beta\varepsilon_F} - 1)$$

[5 marks]

(c) Calculate the grand potential function and find the ratio PV/U where P, V, and U are the pressure, volume and internal energy.

Solution

The grand potential is

$$\mathcal{G} = -k_B T \sum_i g_i \ln(z\mathrm{e}^{-\beta\varepsilon_i} + 1)$$

Similarly, $\sum_i = A\int \frac{d^2k}{(2\pi)^2}$ and $g_i = g$, the grand potential is

$$\mathcal{G} = -k_B T \frac{Ag}{2\pi}\int kdk \ln(z\mathrm{e}^{-\beta\varepsilon_k} + 1)$$

Changing of variable, $k = \sqrt{2m\varepsilon}/\hbar$, This gives $dk = \frac{1}{2}\sqrt{2m}/\hbar\varepsilon^{-1/2}d\varepsilon$. We obtain

$$\begin{aligned}\mathcal{G} =& -k_B T \frac{Amg}{2\pi\hbar^2}\int d\varepsilon \ln(z\mathrm{e}^{-\beta\varepsilon} + 1)\\ =& -k_B T \frac{Amg}{2\pi\hbar^2}\Big[\int d\varepsilon \frac{\beta\varepsilon z\mathrm{e}^{-\beta\varepsilon}}{z\mathrm{e}^{-\beta\varepsilon} + 1}\Big]\\ =& -\frac{Amg}{2\pi\hbar^2}\Big[\int d\varepsilon \frac{\varepsilon}{z^{-1}\mathrm{e}^{\beta\varepsilon} + 1}\Big]\end{aligned}$$

where in the last line, we used integrating by part. Change of variable again $x = \beta\varepsilon$. This gives

$$\mathcal{G} = -\frac{Amg}{2\pi\hbar^2\beta^2}\Big[\int dx \frac{x}{z^{-1}\mathrm{e}^{x} + 1}\Big] = -\frac{Amg}{2\pi\hbar^2\beta^2} f_2(z)$$

Using $\mathcal{G} = -PV$ where $V = A$, we have

$$PV = \frac{Amg}{2\pi\hbar^2\beta^2} f_2(z)$$

The total energy can be calculated by using

$$U = -\Big(\frac{\partial \ln \mathcal{Q}}{\partial\beta}\Big)_z = -\Big(\frac{\partial[\frac{Amg}{2\pi\hbar^2\beta} f_2(z)]}{\partial\beta}\Big)_z = \frac{Amg}{2\pi\hbar^2\beta^2} f_2(z)$$

Therefore, we have

$$\frac{PV}{U} = 1$$

[5 marks]

(d) What is the heat capacity C_V compare to the 3D case.

Solution

The internal energy is

$$U = \frac{Amg}{2\pi\hbar^2\beta^2}\int dx \frac{x}{z^{-1}\mathrm{e}^x + 1} = \frac{Amg}{2\pi\hbar^2}\int d\varepsilon \frac{\varepsilon}{\mathrm{e}^{\beta(\varepsilon-\mu)}+1}$$

Using the Sommerfeld expansion, we have

$$\begin{aligned} U &\approx \frac{Amg}{2\pi\hbar^2}\left[\int_0^\mu \varepsilon d\varepsilon + \frac{\pi^2}{6}\frac{1}{\beta^2} + \dots\right] \\ \frac{U}{N} &\approx \frac{1}{\varepsilon_F}\left[\int_0^\mu \varepsilon d\varepsilon + \frac{\pi^2}{6}\frac{1}{\beta^2} + \dots\right] \\ \frac{U}{N} &\approx \frac{1}{\varepsilon_F}\left[\frac{1}{2}\mu^2 + \frac{\pi^2}{6}\frac{1}{\beta^2} + \dots\right] \end{aligned}$$

In low temperature, we may let $\mu \approx \varepsilon_F$. We obtain

$$\frac{U}{N} \approx \frac{\varepsilon_F}{2}\left[1 + \frac{\pi^2}{3}\frac{k_B^2T^2}{\varepsilon_F^2} + \dots\right]$$

Keeping up to the quadratic term, the heat capacity is

$$C_V = \frac{\partial U}{\partial T} = \frac{\partial}{\partial T}\frac{\varepsilon_F}{2}\left[1 + \frac{\pi^2}{3}\frac{k_B^2T^2}{\varepsilon_F^2}\right] = \frac{Nk_B\pi^2}{3}\frac{k_BT}{\varepsilon_F}$$

Compare to 3D, it has the same linear behavior in T. **[5 marks]**

(e) Calculate the probability for a particle to *occupy* the state with energy $\varepsilon = \mu + \delta$ (above chemical potential) and the probability to *unoccupy* the states with energy $\varepsilon = \mu - \delta$ (below chemical potential). Do these two probability the same?

Solution

From Problem 2 (c), the probability to have n_ε particles in state ε is

$$p(n_\varepsilon) = \frac{\mathrm{e}^{-\beta(\varepsilon-\mu)n_\varepsilon}}{1+\mathrm{e}^{-\beta(\varepsilon-\mu)}}$$

Therefore, the probability to occupy $\varepsilon+\mu$ is

$$p_{\mu+\delta}(1) = \frac{\mathrm{e}^{-\beta\delta}}{1+\mathrm{e}^{-\beta\delta}}$$

The probability for unoccupy $\mu-\varepsilon$

$$p_{\mu-\delta}(0) = \frac{\mathrm{e}^{0}}{1+\mathrm{e}^{\beta\delta}} = \frac{1}{1+\mathrm{e}^{\beta\delta}}$$

Their probability are the same.**[5 marks]**